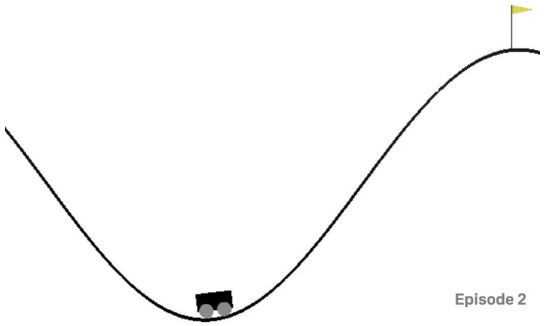
HW1: Implement DQN in MountainCar-v0

observation:

Num	Observation	Min	Max
0	Car Position	-1.2	0.6
1	Car Velocity	-0.07	0.07



action:

Num	Action
0	Accelerate to the Left
1	Don't accelerate
2	Accelerate to the Right

reward

Reward of 0 is awarded if the agent reached the flag (position = 0.5) on top of the mountain.

Reward of -1 is awarded if the position of the agent is less than 0.5.

Code Demo

```python
import gym

env = gym.make('MountainCar-v0')
action_space = env.action_space.n
state_space = env.observation_space.shape[0]

while True:

    state = env.reset()
    terminated = False

    while terminated == False:

        action = my_policy(state) #epsilon-greedy
        next_state, reward, terminated, _ = env.step(action)
        buff.add(state, action, reward, next_state, terminated)
        state = next_state


    '''
    Sample a batch from buff
    Train the policy
    '''
```

Requirements

Install gym (https://github.com/openai/gym)

Implement DQN algorithm in MountainCar-v0 environment

Draw reward plot (reward versus episode) for your algorithm

Pack your report and code in a zip file, submit it to Blackboard before 2022/3/31 23:59pm.

Tips

The reward in MountainCar-v0 is relatively sparse, if simple epsilon greedy does not work, you can try some special exploration method, for example:

HER(https://arxiv.org/abs/1707.01495)

RND(https://arxiv.org/abs/1810.12894).